

A Validated Isolated Exchange-Model Audit: Reachable Demand, Fixed-Contrast Failure, and a Scoped No-Go

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Abstract

The joint first-alias certificate requires a boundary observation, a fixed reference, a shell scale and contrast, every Duhamel weight, and a far remainder from one coherent data source. We freeze a graded finite-dimensional source defined from exact rational data and audit it without fitting any field after the demand is known. Two-state exchange blocks give $\text{Tr } K_c^m = 1 + c^m$; diagonal alias blocks and scalar parity blocks realize the other packets exactly. The boundary, observation error, Duhamel defects, and far remainder vanish in this model. Nevertheless its frozen contrast $4/5$ fails the joint equation. We prove

$$\frac{\mathcal{P}_k}{\mathcal{A}_k} \longrightarrow C_* C_M \in (0, 1), \quad \frac{e_k}{\mathcal{A}_k} \longrightarrow -(1 - C_* C_M) < 0, \quad \frac{e_k}{kR^{-2k}} \longrightarrow -\infty.$$

The required even power remains eventually reachable, but its contrast radius tends to one. Thus zero best-case reachability does not imply matching for this pre-frozen isolated model. The family is graded rather than one all-order operator, and it is not identified with the actual noisy transfer operator. No full-trace replacement, full-trace divergence, or Riemann-hypothesis conclusion follows.

1 Frozen data and the audit question

All constants in this paper are model definitions, and the audit domain is the set of integers $k \geq 2$. Put

$$r_H = \frac{17}{20}, \quad R = \frac{7}{5}, \quad \Lambda = \frac{2098216888035403}{1250000000000000}, \quad (1)$$

$$C_M = \frac{9731714526004839}{5000000000000000}, \quad C_* = \frac{26314633984227}{2500000000000000}, \quad C_b = \frac{1152012873}{25000000000}. \quad (2)$$

The terminating decimal strings from which these fractions arise are treated as exact rationals. They are not interval enclosures of physical constants. We also freeze

$$c_{\text{iso}} = \frac{4}{5}, \quad c_0 = \frac{3}{5}, \quad c_b = \frac{1}{2}, \quad \eta = 0, \quad \sigma_k = \Lambda^{-2k}. \quad (3)$$

The clearance metadata are C_b and the inherited orientation $(+, -, +)$. Because the boundary blocks below are identical, these metadata do not enter a fitted correction.

Define

$$\beta^2 = \frac{1}{r_H^2 \Lambda}, \quad a_k = \frac{2k-2}{C_M} + 2, \quad \mathcal{A}_k = a_k \beta^{2k}, \quad H_k = kR^{-2k}. \quad (4)$$

The RH-328 fixed-reference equation asks whether

$$e_k = \mathcal{B}_k + \mathcal{S}_k + \mathcal{R}_k + \mathcal{P}_k - \mathcal{A}_k = o(H_k) \quad (5)$$

for data obtained before inspecting the demand. Our answer is negative for the model defined below.

Remark 1.1 (Graded scope). For each k we define a finite-dimensional collection of blocks. Their dimensions and some roots depend on k . We do not claim that one operator has all these power traces simultaneously. This distinction is part of the theorem statement, not a technical omission.

2 Exact block realization

For $0 \leq c \leq 1$ let

$$K_c = \frac{1}{2} \begin{pmatrix} 1+c & 1-c \\ 1-c & 1+c \end{pmatrix}. \quad (6)$$

The symmetric and antisymmetric unit vectors are eigenvectors with eigenvalues 1 and c .

The positive contrast range is a deliberate model restriction. We do not claim to identify the sign in the larger even-power class $|c| \leq 1$.

Lemma 2.1 (Exchange trace law). *For every integer $m \geq 1$,*

$$\text{Tr } K_c^m = 1 + c^m. \quad (7)$$

Proof. Diagonalize K_c in its symmetric–antisymmetric basis and take the trace. $\square$

Let $\beta_k > 0$ be the unique root satisfying

$$\beta_k^{2k} = \frac{\beta^{2k}}{C_M} \quad (8)$$

and define the $2k$ -dimensional alias block

$$C_k = \text{diag}(\beta_k I_{2k-2}, \beta I_2). \quad (9)$$

For the parity channel put

$$\delta_k = C_* \Lambda^{-k}, \quad J_k^{\text{ref}} = [r_H^{-1}], \quad J_k^{\text{iso}} = [r_H^{-1}(1 - \delta_k)]. \quad (10)$$

Finally set $s_k = \mathcal{A}_k^{1/(2k)}$ and define

$$D_k^{\text{iso}} = D_k^{\text{ref}} = s_k K_{c_b}, \quad (11)$$

$$Q_k^{\text{iso}} = s_k K_{c_{\text{iso}}}, \quad Q_k^{\text{ref}} = s_k K_{c_0}. \quad (12)$$

Proposition 2.2 (Operator-coherent packet identities). *The blocks above give*

$$\text{Tr } C_k^{2k} = \left(\frac{2k-2}{C_M} + 2 \right) \beta^{2k} = \mathcal{A}_k, \quad (13)$$

$$\mathcal{P}_k := \text{Tr}\{(J_k^{\text{ref}})^{2k} - (J_k^{\text{iso}})^{2k}\} = r_H^{-2k} \{1 - (1 - \delta_k)^{2k}\}, \quad (14)$$

$$\mathcal{B}_k := \text{Tr}\{(D_k^{\text{iso}})^{2k} - (D_k^{\text{ref}})^{2k}\} = 0, \quad (15)$$

$$\mathcal{S}_k := \text{Tr}\{(Q_k^{\text{iso}})^{2k} - (Q_k^{\text{ref}})^{2k}\} = \mathcal{A}_k(c_{\text{iso}}^{2k} - c_0^{2k}). \quad (16)$$

There is no omitted block, so $\mathcal{R}_k = 0$, and the matrix observation is the definition of the shell, so $\mathcal{E}_k^{\text{obs}} = 0$.

Proof. Equation (8) and the multiplicities in (9) give (13). The scalar power difference gives (14). Identical boundary blocks give (15). Apply (7) to the two scaled shell blocks; the common eigenvalue-one contribution cancels and $s_k^{2k} = \mathcal{A}_k$, proving (16). $\square$

Every scale and contrast in this proposition is frozen by (1)–(3). In particular, we do not solve the matching equation for c_{iso} and then insert the result into the model.

3 Exact matching equation and asymptotic no-go

In this model the signed demand and required even power are

$$D_k = \mathcal{A}_k - \mathcal{P}_k - \mathcal{B}_k = \mathcal{A}_k - \mathcal{P}_k, \quad (17)$$

$$y_k = c_0^{2k} + \frac{D_k}{\mathcal{A}_k} = c_0^{2k} + 1 - \frac{\mathcal{P}_k}{\mathcal{A}_k}. \quad (18)$$

The best-case RH-328 reachability screen is zero precisely when $y_k \in [0, 1]$. The actual fixed-contrast mismatch is different:

$$\frac{e_k}{\mathcal{A}_k} = c_{\text{iso}}^{2k} - y_k = c_{\text{iso}}^{2k} - c_0^{2k} + \frac{\mathcal{P}_k}{\mathcal{A}_k} - 1. \quad (19)$$

Lemma 3.1 (Small parity gap). *If $x_k = C_*\Lambda^{-k}$, then*

$$\frac{1 - (1 - x_k)^{2k}}{2kx_k} \rightarrow 1. \quad (20)$$

Proof. Here $x_k \rightarrow 0$ and $kx_k \rightarrow 0$. The inequalities $1 - (1-x)^n \leq nx$ and $1 - (1-x)^n \geq nx - \binom{n}{2}x^2$ for $0 \leq x \leq 1$ squeeze the ratio between $1 - (2k-1)x_k/2$ and 1 for all sufficiently large k . $\square$

Theorem 3.2 (Validated isolated fixed-contrast failure). *For the frozen graded family,*

$$\frac{\mathcal{P}_k}{\mathcal{A}_k} \longrightarrow \gamma := C_* C_M, \quad 0 < \gamma < 1, \quad (21)$$

$$y_k \longrightarrow 1 - \gamma \in (0, 1), \quad (22)$$

$$\frac{e_k}{\mathcal{A}_k} \longrightarrow -(1 - \gamma) < 0, \quad (23)$$

$$\frac{\mathcal{A}_k}{H_k} \longrightarrow +\infty, \quad \frac{e_k}{H_k} \longrightarrow -\infty. \quad (24)$$

Thus the reachability screen is eventually exactly zero, while the frozen model fails the target matching condition. If $r_k = y_k^{1/(2k)}$ is the required contrast radius, then

$$r_k \longrightarrow 1, \quad r_k - c_{\text{iso}} \longrightarrow \frac{1}{5}. \quad (25)$$

Moreover, the RH-328 precision scales are exactly

$$\frac{H_k}{\mathcal{A}_k} = \frac{k}{a_k}(\beta R)^{-2k}, \quad \frac{H_k}{k\mathcal{A}_k} = \frac{1}{a_k}(\beta R)^{-2k}. \quad (26)$$

The power mismatch in (19) is not $o(H_k/\mathcal{A}_k)$, and the radius gap in (25) is not $o(H_k/(k\mathcal{A}_k))$.

Proof. Because $\beta^{2k} = r_H^{-2k} \Lambda^{-k}$, the preceding lemma yields

$$\frac{\mathcal{P}_k}{A_k} = \frac{2kC_*}{a_k} \{1 + o(1)\} \longrightarrow C_* C_M,$$

since $a_k/k \rightarrow 2/C_M$. Exact integer comparison gives

$$\gamma = \frac{256086505790802487295251674453}{125000000000000000000000000} \in (0, 1). \quad (27)$$

Both fixed contrasts are strictly subunit, so (18) and (19) prove (22) and (23). In particular, $y_k \in (0, 1)$ for all sufficiently large k .

Next,

$$\frac{\mathcal{A}_k}{H_k} = \frac{a_k}{k}(\beta R)^{2k}, \quad (\beta R)^2 = \frac{980000000000000000}{606384680642231467} > 1. \quad (28)$$

The first factor tends to $2/C_M > 0$, proving alias-to-target divergence. Multiplying it by the negative nonzero limit in (23) proves $e_k/H_k \rightarrow -\infty$. Finally $\log y_k$ converges to the finite number $\log(1-\gamma)$, so $\log r_k = (\log y_k)/(2k) \rightarrow 0$. Hence $r_k \rightarrow 1$ and (25) follows. Inverting (28) gives the first identity in (26); division by k gives the second. Both right sides tend to zero exponentially, whereas (19) has the nonzero limit $-(1-\gamma)$ and the radius gap tends to $1/5$. This proves both precision failures. $\square$

This theorem is stronger than a finite failed fit: its sign and target-scale failure hold eventually by exact asymptotics. It is weaker than an actual noisy-trace theorem because every block belongs only to the isolated model.

4 Duhamel and omitted-remainder audit

The product comparison is deliberately recorded leg by leg even though the model representation is exact. In each shell channel $c \in \{c_{\text{iso}}, c_0\}$, compare the $2k$ factors

$$A_{j,k}^{(c)} = G_{j,k}^{(c)} = Q_{c,k}, \quad 1 \leq j \leq 2k. \quad (29)$$

For the spectral matrix norm, $\|K_c\| = 1$ and hence $\|Q_{c,k}\| = s_k$. The trace observation on two-dimensional matrices obeys $|\text{Tr } X| \leq 2\|X\|$. Therefore every telescoping weight is

$$W_{j,k}^{(c)} = 2s_k^{2k-1} = 2\mathcal{A}_k^{(2k-1)/(2k)}, \quad (30)$$

while every leg defect is $\delta_{j,k}^{(c)} = \|A_{j,k}^{(c)} - G_{j,k}^{(c)}\| = 0$.

Proposition 4.1 (Complete isolated Duhamel ledger). *Across the noisy and reference shell channels there are exactly $4k$ prefix/suffix weights. None is discarded, and*

$$\mathcal{U}_k = \sum_{c \in \{c_{\text{iso}}, c_0\}} \sum_{j=1}^{2k} W_{j,k}^{(c)} \delta_{j,k}^{(c)} = 0. \quad (31)$$

Together with $\mathcal{E}_k^{\text{obs}} = \mathcal{R}_k = 0$, the best- and worst-case uncertainty residuals both equal $|e_k|$.

The zero in (31) certifies internal implementation of the isolated block identity. It does not bound the defect between an actual noisy critical leg and this model. Likewise $\mathcal{R}_k = 0$ says that the model has no omitted block; it is not an estimate for the physical far region.

5 Deterministic experiment protocol

The executable ledger uses Python rational arithmetic for all constants, packets, signs, reachability decisions, and target-window verdicts. Decimal endpoints are rounded outward only for display. Matrix multiplication and powers independently check (7). The selected orders are $k = 2, 4, 8, 16, 24, 32$; across them the ledger retains 344 Duhamel weights. The following subset is printed compactly.

k	$\mathcal{P}_k/\mathcal{A}_k$	y_k	$e_k/\mathcal{A}_k$	$ e_k /H_k$
2	0.131466	0.998134	-0.588534	2.32697
4	0.158186	0.858610	-0.690838	5.98856
8	0.180921	0.819361	-0.791213	42.3139
16	0.193349	0.806651	-0.805859	1899.64
32	0.198985	0.801015	-0.801015	3.97557×10^6

Every displayed row has $y_k \in [0, 1]$ and $|e_k| > H_k$, with both decisions made before decimal conversion. These rows reproduce formulas; they are not used to infer any limit in theorem 3.2.

6 Boundary and RH-330 handoff

The validated conclusion is precisely this: the pre-frozen graded isolated exchange family fails the RH-328 joint matching equation, even though its best-case contrast reachability screen eventually passes. The conclusion does not identify the family with the actual noisy operator, establish a physical shell scale or contrast, control actual moving-order Duhamel defects, prove an actual far-remainder little- o estimate, replace the full trace, or prove actual full-trace divergence.

RH-330 must state a transfer criterion retaining boundary, shell, parity, alias, observation, and far-remainder terms on one clock. It must quantify the replacement of every isolated field and preserve signed cancellation at scale $o(H_k)$. Passing or failing a different isolated model cannot bypass that transfer step.

All five program gates remain false/open: no canonical intrinsic dynamical spectral determinant, oriented unitary completion, self-adjoint generator with an intrinsic $T \log T$ law, von Mangoldt prime-power trace, or equality with the completed-zeta divisor is proved. In particular, no Hilbert–Polya operator is constructed, no Riemann zero is identified, and the Riemann Hypothesis is not proved.